(1) Let,

$A = \{$Customer Buys more than one item$\}$

$B = \{$Customer Buys a blue shirt$\}$.

$\therefore P(A) = 0.6$; $P(A') = 0.4$

$P(B) = 0.3$. ; $P(B') = 0.7$

$P(A \cap B) = 0.1$;

$\therefore$ Now,

$$P(A' \cap B') = P(A') + P(B') - P(A' \cup B')$$
$$= P(A') + P(B') - P(A \cap B)'$$
$$= P(A') + P(B') - [1 - P(A \cap B)]$$
$$= 0.4 + 0.7 - [1 - 0.1]$$
$$= 0.2$$

~~(2) $S = \{(H,1), (H,2), (H,3), (H,4), (H,5), (T,1), (T,2), (T,3), (T,4), (T,5)\}$~~

~~$\therefore P(A) = \frac{7}{10}$~~

P.T.O

②

② $S = \{(H,1), (H,2), (H,3), (H,4), (H,5)$
$(T,1), (T,2), (T,3), (T,4), (T,5)\}$

$P(A) = 2/10$

$A = \{$ ~~No of~~ Head$\}$
$B = \{2 \text{ or } 4\}$
$n(A) = 5$; $n(B) = 4$
$n(S) = 10$
$n(C \cap B) = 2$
$\therefore P(A) = \frac{n(C \cap B)}{n(S)} = 2/10$

③ Given that,
$P(A) = 0.38$; $P(A') = 0.62$
$P(B) = 0.42$; $P(B') = 0.58$

If A and B are mutualley exclusive then,
$P(A \cap B) = 0.$

$\therefore P(A' \cup B') = P(A \cap B)'$
$= 1 - P(A \cap B)$
$= 1 - 0$
$= 1.$

And if they are mutually indepentent then,
$P(A \cap B) = P(A) \times P(B) = \cancel{1/4}\ 0.38 \times 0.42 = 0.1596$
~~$\therefore P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B) = 1 - 0.1596$~~
$\therefore P(A' \cap B') = P(A') \times P(B') = 0.3596$ ~~$= 0.8404$~~

③

④

	$n(A_1)$	$n(A_2)$	Total
$n(B_1)$	25	40	65
$n(B_2)$	55	30	85
Total	80	70	150

$$P(A_2) = \frac{n(A_2)}{n(S)} = \frac{70}{150}$$

$$P(A_1|B_2) = \frac{n(A_1 \cap B_2)}{n(B_2)} = \frac{30}{85}$$

$$P(B_2|A_1) = \frac{n(A_1 \cap B_2)}{n(A_1)} = \frac{55}{80}$$

$$P(A_2 \cap B_1) = \frac{n(A_2 \cap B_1)}{n(S)} = \frac{40}{150}$$

$$P(A_1 \cap B_2) = \frac{n(A_1 \cap B_2)}{n(S)} = \frac{55}{150}$$

$$P(A_1 \cup B_2) = P(A_1) + P(B_2) - P(A_1 \cap B_2)$$

$$= \frac{80}{150} + \frac{85}{150} - \frac{55}{150}$$

$$= \frac{110}{150}$$

④ ⑤ $S = \{ \boxed{(H,1)}, (H,2), \boxed{(H,3)}, (H,4)$
$(T,1), (T,2), (T,3), (T,4) \}$

$\frac{4}{8} = \frac{n(A)}{n(S)} = P(A) \cancel{= 2/4}$; $A = \{$odd sides in the dice$\}$

$\frac{4}{8} = \frac{n(B)}{n(S)} = P(B) \cancel{= 1/2}$; $B = \{$head in coin$\}$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{8} = 1/4$$

∴ Check is independent or conditional,

$$P(A \cap B) = P(A) \times P(B) = 4/8 \times 4/8$$
$$\cancel{= 2/4 \times 1/4} = 1/4$$
$$\cancel{= 1/8}$$

∴ A and B are independent.

If A and B are mutually exclusive,

$$P(A \cap B) = 0; P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1$$

$$P(A' \cap B') = P(A \cup B)'$$
$$= 1 - P(A \cup B)$$
$$= 1 - 1$$
$$= 0$$

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6

Sample Space = {

(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),

(2,1), (2,2), (2,3), (2,4), (2,5), $\boxed{(2,6)}$

(3,1), (3,2), (3,3), $\boxed{(3,4)}$, (3,5), (3,6)

(4,1), (4,2), $\boxed{(4,3)}$, (4,4), (4,5), (4,6)

(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)

(6,1), $\boxed{(6,2)}$, (6,3), (6,4), (6,5), (6,6) }

$$P(R) = \frac{n(R)}{n(\text{Sample space})} = \frac{n(R)}{n(\text{Sample space})} = \frac{6}{36}$$

$$P(S) = \frac{n(S)}{n(\text{Sample space})} = \frac{4}{36}$$

$$\therefore P(R \cap S) = \frac{n(R \cap S)}{n(S)} = \frac{2}{36}$$

$\therefore$ If independent,

$$P(R \cap S) = P(R) \times P(S)$$
$$= 6/36 \times 4/36$$
$$= 1/54$$
$$\neq P(R \cap S)$$

$\therefore$ R and S are not independent.

7 Given that,

$P(I) = 0.7$ | $P(I') = 0.3$

$P(II) = 0.8$ | $P(II') = 0.2$

$P(III) = 0.6$ | $P(III') = 0.4$

i) $P(Q) = P(I') \times P(II') \times P(III')$; Q – {Task will not be completed}

$= 0.024$

ii) $P(R) = P(I) \times P(II) \times P(III)$; R = {every worker will complete the task}

$= 0.336$.

iii) S = {only the second worker will complete the task.}

$P(S) = P(I') \times P(II) \times P(III')$

$= 0.096$

8 $P(I) = 0.35$; $P(II) = 0.4$; $P(III) = 0.25$

$P(B_{1/I}) = 1 - 0.97 = 0.03$; $P(B_{2/II}) = 1 - 0.92 = 0.08$

$P(B_{3/III}) = 1 - 0.95 = 0.05$

$\therefore P(B) = P(I)P(B_{1/I}) + P(II)P(B_{2/II}) + P(III)P(B_{3/III})$

$=$

7

8 $P(I) = 0.35$; $P(II) = 0.~~3~~40$; $P(III) = 0.25$

(Probabilities of non-efficiency) $P(B/I) = 1 - 0.92 = 0.08$

$P(B/II) = 1 - 0.97 = 0.03$

$P(B/III) = 1 - 0.95 = 0.05$

$\therefore P(B) = P(I)\, P(B/I) + P(II)\, P(B/II) + P(III)\, P(B/III)$

$= 0.0525$

$\therefore$ ~~Let,~~ $P(II/B) = \dfrac{P(II)\, P(B/II)}{P(B)}$

$\cancel{= 0.02} = \dfrac{0.4 \times 0.03}{0.0525}$

$= 0.2285$

9 $P(T^+/D^-) = 0.12$; $P(T^-/D^-) = 0.88$

$P(T^-/D^+) = 0.09$; $P(T^+/D^+) = 0.91$

$P(D^+) = 0.05$; $P(D^-) = 0.95$

$\therefore P(D^+/T^-) = \dfrac{P(D^+)\, P(T^-/D^+)}{P(D^+)\, P(T^-/D^+) + \cancel{P(D^+)\, P(D^{T+}/D^+)}\; P(D^-)\, P(T^-/D^-)}$

$= 0.54$

8

10 Given that,

$E[X] = -0.05$

$\Rightarrow -0.4 - a + 0 + 0.3 + 2b = -0.05$

$\therefore 2b - a = 0.05$ — (I)

Again,

$\Sigma f(x) = 1$

$\Rightarrow 0.2 + a + 0.1 + 0.3 + b = 1$

$\Rightarrow a + b = 0.4$ — (II)

~~$\therefore$ (II) — (I),~~

~~$a + b - (2b - a) = 0.35$~~

~~$\Rightarrow a + b - 2b + a = 0.35$~~

~~$\therefore$ (I) — (II),~~

~~$2b - a - (a+b) = -0.35$~~

~~$\Rightarrow b = -0.35$ — (III)~~

~~$\therefore$ (II),~~

~~$a = 0.4 - b$~~

~~$\therefore a = 0.75$~~

$\therefore$ (II) + (I),

$a + b + 2b - a = 0.45$

$\Rightarrow 3b = 0.45$

$\therefore b = 0.15$

$\therefore$ (II),

$a = 0.25$

$\therefore \left\{ \begin{array}{l} a = 0.25 \\ b = 0.15 \end{array} \right.$ Ans.

(ii) Given that,

$$E[X] = -0.05$$

$$\Rightarrow \quad -0.4 - 0.25 + 0.1a + b + 0.3 = -0.05$$

$$\Rightarrow \quad 0.1a + b = 0.3 \quad - (i)$$

Also,

$$\Sigma f(x) = 1$$

$$\Rightarrow \quad 0.2 + 0.25 + 0.1 + b + 0.15 = 1$$

$$\therefore b = 0.3 \quad - (ii)$$

Now

(i),

$$0.1a + b = 0.3$$

$$\Rightarrow 0.1a = 0.3 - b$$

$$\Rightarrow 0.1a = 0.3 - 0.3$$

$$\Rightarrow 0.1a = 0$$

$$\therefore a = 0.$$

$$\therefore \quad a = 0$$
$$b = 0.3$$

(12) $E[X] = -0.05$

$\Rightarrow 0.2a - 0.25 + 0 + 0.3b + 0.3 = -0.05$

$\Rightarrow 0.2a + 0.3b = \cancel{-0.05} - 0.1$

$\Rightarrow a = -0.5 - 1.5b$ — (1)

$E[X^2] = 1.95$

$\Rightarrow 0.2a^2 + 0.25 + 0 + 0.3b^2 + 0.6 = 1.95$

$\Rightarrow 0.2a^2 + 0.3b^2 = 1.95 - 0.85$

$\Rightarrow 0.2a^2 + 0.3b^2 = 1.1$

$\Rightarrow 0.2(-0.5 - 1.5b)^2 + 0.3b^2 = 1.1$

$\Rightarrow 0.2(0.25 + 1.5b + b^2 2.25) + 0.3b^2 = 1.1$

$\Rightarrow 0.05 + 0.3b + 0.75b^2 = 1.1$

$\Rightarrow 0.75b^2 + 0.3b - 1.05 = 0$

$\therefore b = 1, -1.4$

For $b = 1$ in (1),	For $b = -1.4$ in (1)
$a = -0.5 - 1.5b$	$a = -0.5 - 1.5b$
$\therefore a = -2$	$\therefore a = 1.6$

We accept the values $a = -2$ and $b = 1$ as they are countable values in discrete random variables.

13

a) pmf,

$$f(x) = \frac{y_{ext} - |x - x_{ext}|}{m}$$

$$= \frac{3 - |x-4|}{9}$$

b) mean,

$$E[X] = \mu = \sum x f(x)$$

$$= 2 \times \frac{1}{9} + 3 \times \frac{2}{9} + 4 \times \frac{3}{9} + 5 \times \frac{2}{9} + 6 \times \frac{1}{9}$$

$$= 4.$$

c) $f(2) = 1/9$; $f(3) = 2/9$; $f(4) = 3/9$; $f(5)$ is $2/9$; $f(6) = 1/9$.

Here, $f(4) = 3/9$ is the maximum,

So, mode is 4.

~~d)~~ We know, $F(x) = P(X \leq x)$

$\therefore F(2) = 1/9$; $F(3) = 3/9$; $F(4) = 6/9$;

$F(5) = 8/9$; $F(6) = 1$.

For median, $F(X) \geq 0.5$, so the X's lowest value will be median. So, the median is 4.

d) $M(t) = E[e^{tx}] = \sum e^{tx} f(x)$

$$= \frac{1}{9}e^{2t} + \frac{2}{9}e^{3t} + \frac{3}{9}e^{4t} + \frac{2}{9}e^{5t} + \frac{1}{9}e^{6t}$$

14. Given that,

$N = 50$

$N_1 = 5$; $N_2 = 45$

$n = 5$

(i) $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$$= \frac{{}^{5}C_0 \times {}^{45}C_5}{{}^{50}C_5} + \frac{{}^{5}C_1 \times {}^{45}C_4}{{}^{50}C_5} + \frac{{}^{5}C_2 \times {}^{45}C_3}{{}^{50}C_5}$$

$= 0.9952$

(ii) $P(X=5) = \dfrac{{}^{5}C_5 \times {}^{45}C_0}{{}^{50}C_5}$

$= 4.72 \times 10^{-7}$.

15. $E[x^2+5] = \sum_{x=1}^{4} (x^2+5) f(x)$

$= \frac{1}{10} \times 6 + \frac{2}{10} \times 9 + \frac{3}{10} \times 14 + \frac{4}{10} \times 21$

$= 15$

$E[(1-3x)] = \sum_{x=1}^{4} (1-3x) f(x)$

$= \frac{1}{10} \times -2 + \frac{2}{10} \times -5 + \frac{3}{10} \times -8 + \frac{4}{10} \times -11$

$= -8$

$$E\left[(1-3x)^2\right] = \sum_{x=1}^{4} (1-3x)^2 f(x)$$

$$= \frac{1}{10} \times 4 + \frac{2}{10} \times 25 + \frac{3}{10} \times 64 + \frac{4}{10} \times 121$$

$$= 73$$

$$\therefore V(1-3x) = E\left[(1-3x)^2\right] - \left\{E[1-3x]\right\}^2$$

$$= 73 - (-8)^2$$

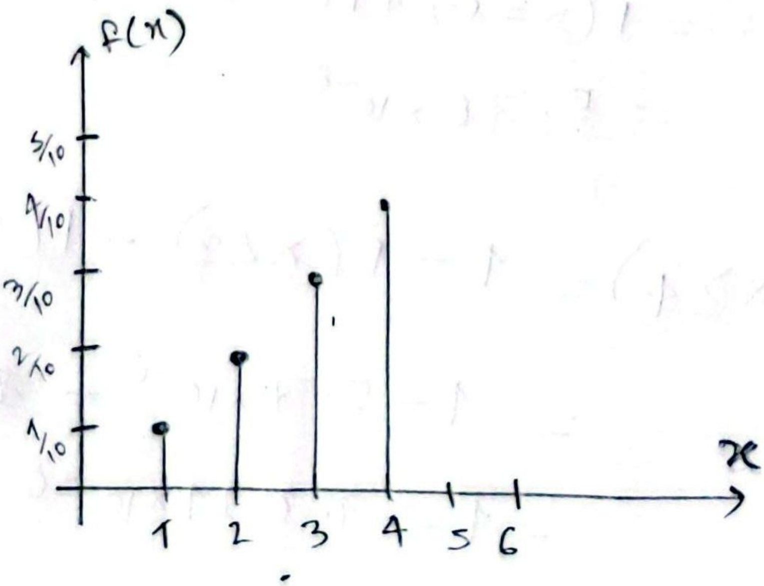
$$= 73 - 64$$

$$= 9.$$

Now,

$P(X=1) = 1/10$; $P(X=2) = 2/10$; $P(X=3) = 3/10$; $P(X=4) = 4/10$

$\therefore$ The line graph:



14

~~(15) Let,~~

~~$N = 100$~~

~~$N_1 = 80$, $N_2 = 20$~~

~~$n = 15$~~

~~$\therefore$ X is distributed,~~

~~$f(x) = {}^{15}C$~~

(16) Given that,

$p = 0.8$

$q = 0.2$

$n = 15$

$$\therefore f(x) = {}^{n}C_{x}(p)^{x}(q)^{n-x}$$

$$= {}^{15}C_{x}(p)^{x}(q)^{15-x}$$

So, assuming independence, X is binomially distributed. That is, X is $b(15, 0.8)$.

For X mgf,

$$M(t) = (q + pe^{t})^{n} = (0.2 + 0.8e^{t})^{15}$$

$$\therefore P(X<3) = P(x=0) + P(x=2) + P(x=1)$$

$$= 5.70 \times 10^{-8}$$

$$\therefore P(X \geq 4) = 1 - P(X<3) - P(x=3)$$

$$= 1 - 5.7\times10^{-8} - 9.54\times10^{-7}$$

$$= 1 - 1.011\times10^{-6}$$

$$= 0.9999$$

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(17) Given that,

$n = 10$

$p = 0.4$

$q = 0.6$

pmf, $f(x) = {}^{10}C_x (0.4)^x (0.6)^{10-x}$

$\therefore P(X>2) = 1 - P(X \leq 2)$

$= 1 - [P(X=0) + P(X=1) + P(X=2)]$

$= 1 - \cancel{0.34}\ 0.1672$

$= \cancel{0.6513}\ 0.8327$

$\therefore$ Variance, $\sigma^2 = npq = 10 \times 0.4 \times 0.6 = 2.4$

(18) Given that,

SD, $\sigma = 2$

$\therefore \sigma^2 = \mu = E[X] = 2^2 = 4.$

$\therefore$ pdf, $f(x) = \dfrac{e^{-\lambda} \lambda^x}{x!}$; $\lambda = 4$; $x = 1, 2, 3, \ldots$

$\therefore P(X \geq 1) = 1 - P(X=0)$

$= 1 - \dfrac{e^{-4} \times (4)^0}{0!}$

$= 0.9816$

(19) $P(X=1) = 2P(X=2)$

$$\Rightarrow \cancel{\frac{e^{-\lambda} \times \lambda^{x}}{x!}} \Rightarrow \frac{e^{-\lambda} \times \lambda^{1}}{1!} = 2 \times \frac{e^{-\lambda} \times \lambda^{2}}{2!}$$

$$\Rightarrow e^{-\lambda} = e^{-\lambda} \times \lambda$$

$$\therefore \lambda = 1.$$

$$\therefore P(X=5) = \frac{e^{-1} \times 1^{5}}{5!}$$

$$= 3.065 \times 10^{-3}$$

$$\therefore SD,\ \sigma = \sqrt{\lambda}$$

$$= 1.$$